WSABI
7/30/2025

Off-Ball Value

Quantifying the effect of individual presses in soccer in EFL 2022/2023

Contents

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“I’d give the Ballon d'Or to Mr. Ousmane Dembele... Goals, titles, leadership, defending, how he was pressing.” - Luis Enrique

Existing Work

1

VPEP

Probability of winning the ball in near future. Only uses simple tracking feature. All turnovers are equal

2

exPress

Like VPEP. Uses player positioning as a variable, but Over-extrapolates the effect of moving players.

3

Bekkers 2024

Physics-based. Modeled time-to-intercept using positioning and vectors to evaluate pressure the team is under.

4

BDP and PPDA

Team-metrics, useful but not super advanced. No individual metrics

Project roadmap

Data Prep

Start on World Cup training. Look at pressure indicators, get familiar with covariates

xT Grid

Train xT grid on full season. 5 iterations, starting with xG

cKDE

Train cKDE xT, make smoothed xT grid using Gaussian cox process

Calculate xT

Use xT pre and post press to find the difference and report that

Conclusions

Make summaries across the season on a player level

Current issues/future plans

	Challenge	How we'll respond
1	Control for defender/opposition quality	Regress across the season with indicators for players
2	Control for simultaneous pressing of teammates	Find average effect of 2 players pressing and give them value accordingly
3	Change xT vs VAEP	Change xT is a flawed metric as it does not adequately incorporate opposition chance of scoring. VAEP would do it better.

xT vs VAEP

	xT	VAEP
Calculation	Probability you score in the next 5 actions.	Change in probability of scoring and conceding due to an action
Losing Ball Value	Only probability your team scores. Losing the ball in a worse spot doesn’t affect our xT Grid	Factors in the probability you score and the other team scores no matter what
Difficulty	Generally not too hard	Usually requires blackbox models

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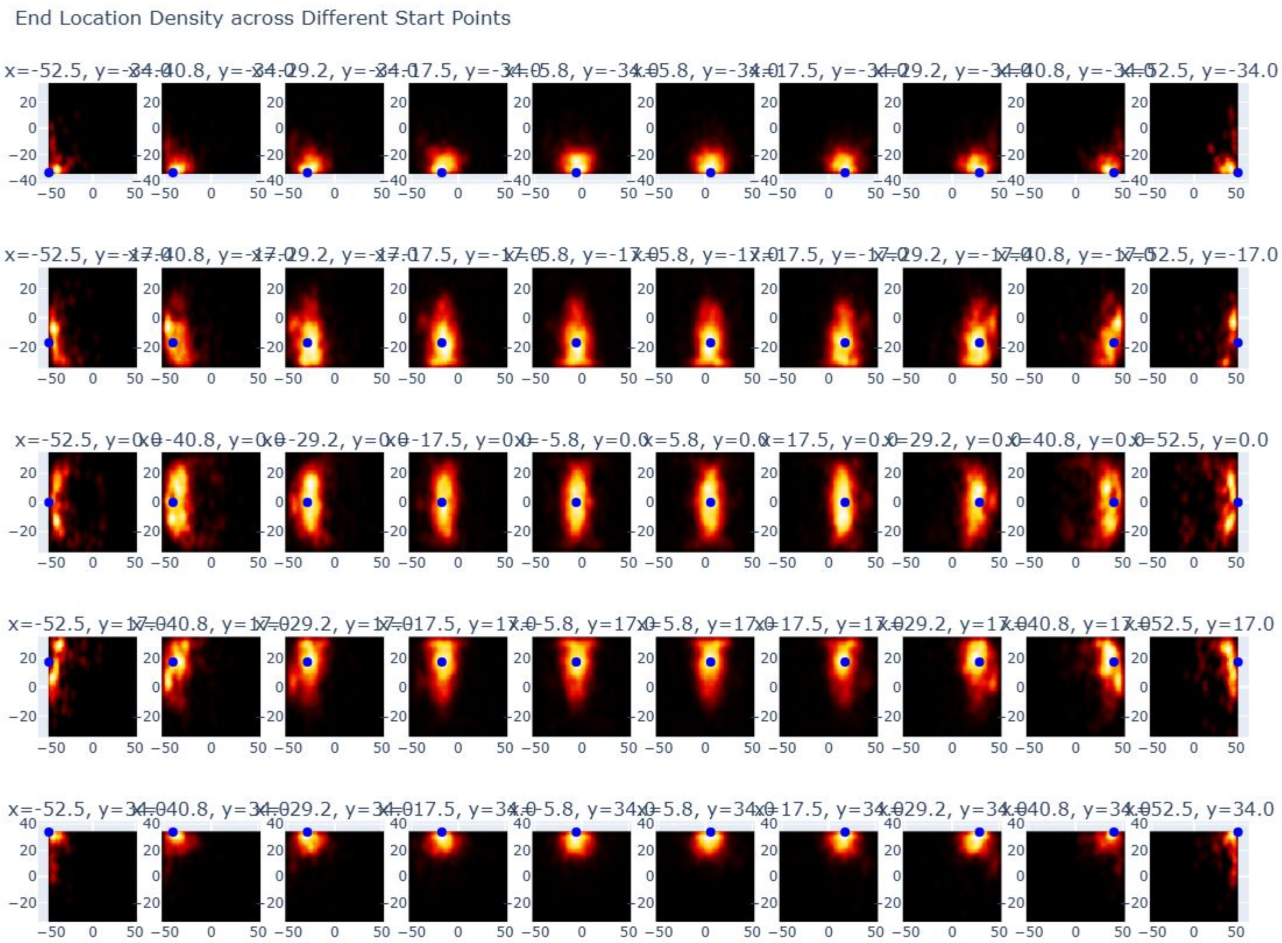
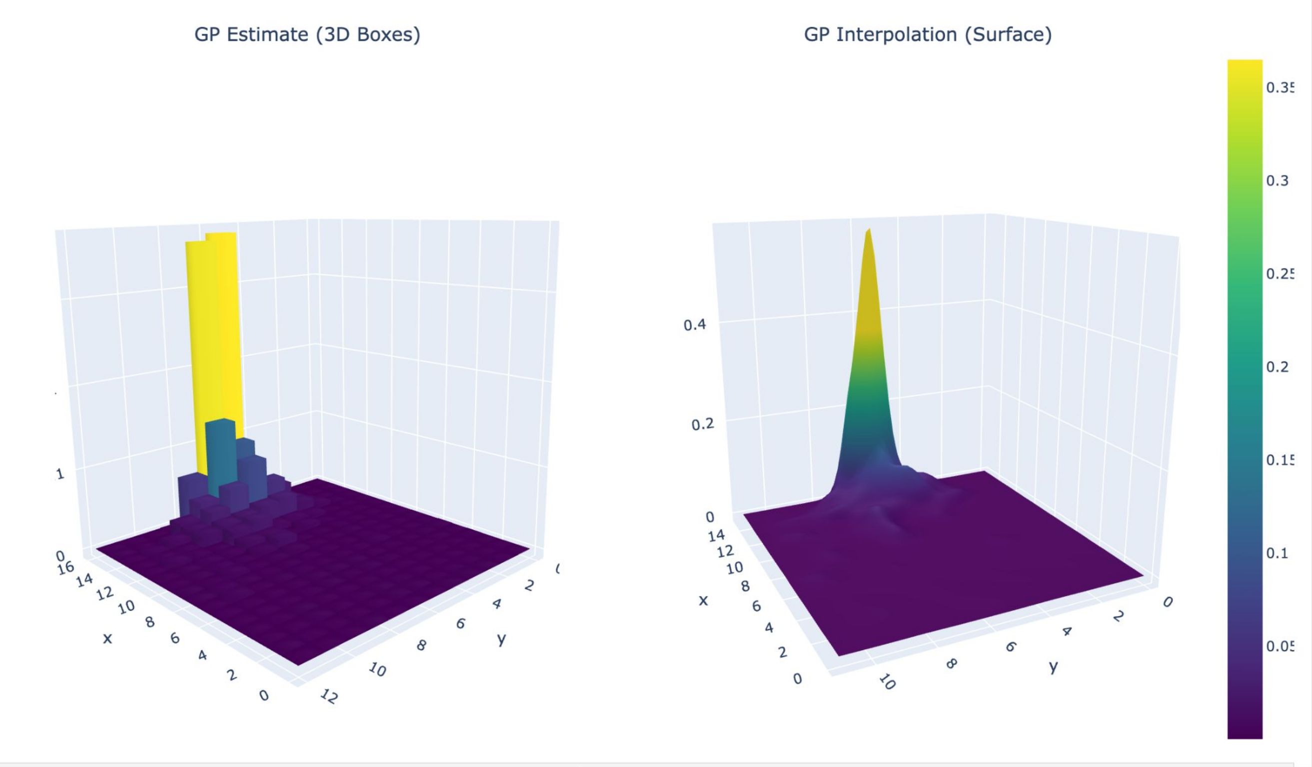
cKDE Plot



xT Grid Comparison



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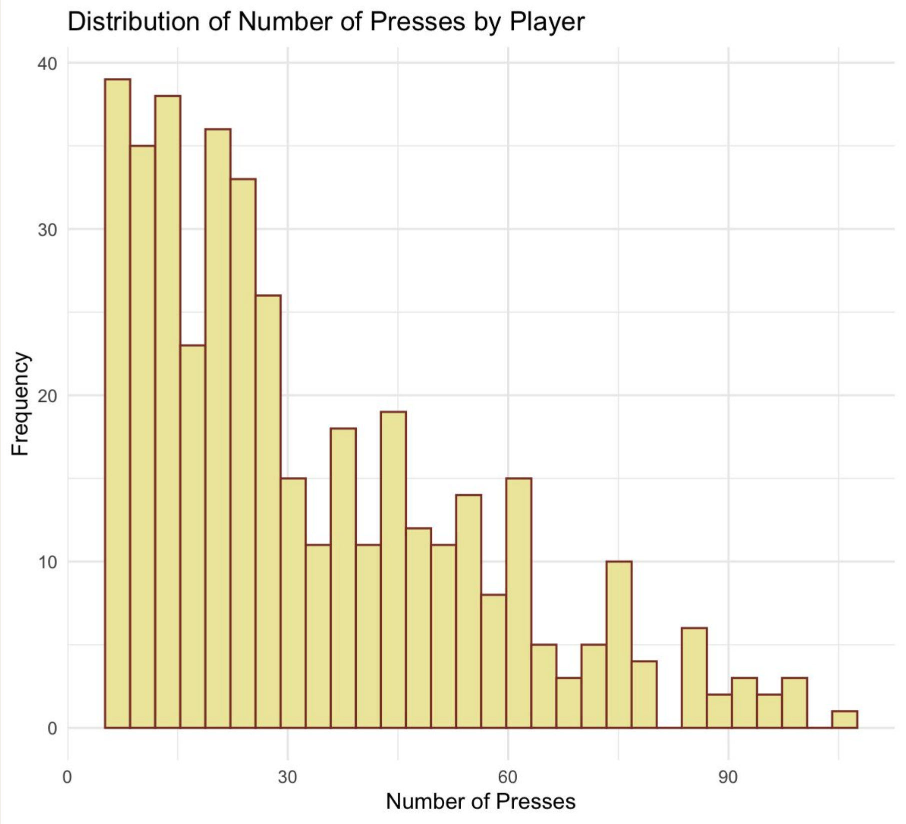
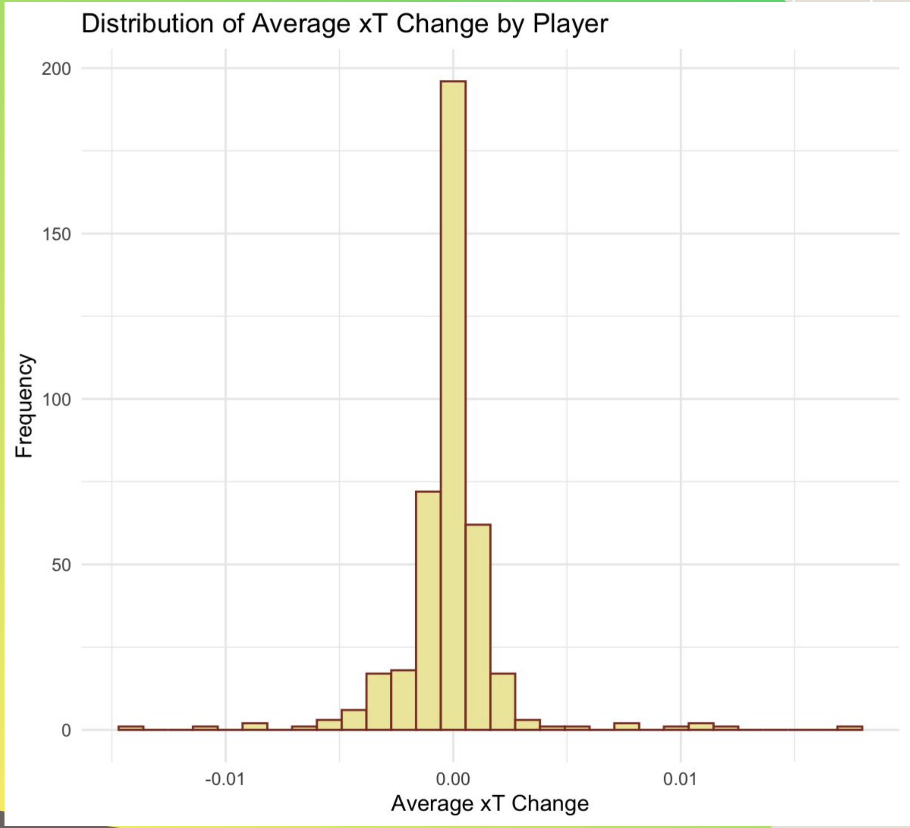


Distribution of Number of Presses by Player
Frequency
40
30
20
10
0
0
30
60
90
Number of Presses



Distribution of Average xT Change by Player
Frequency
200
150
100
50
0
-0.01
0.00
0.01
Average xT Change

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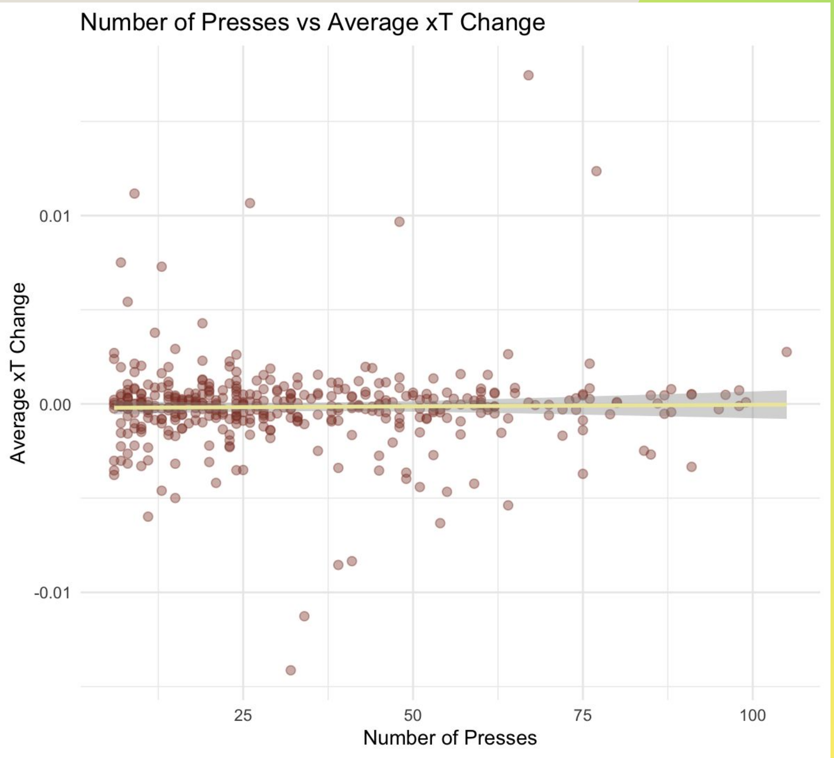
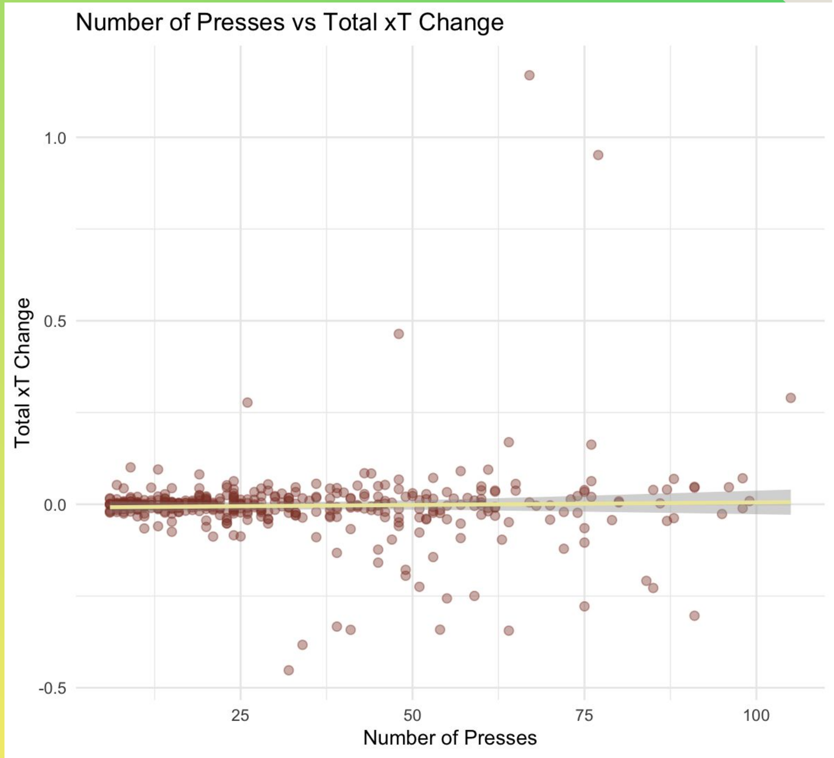
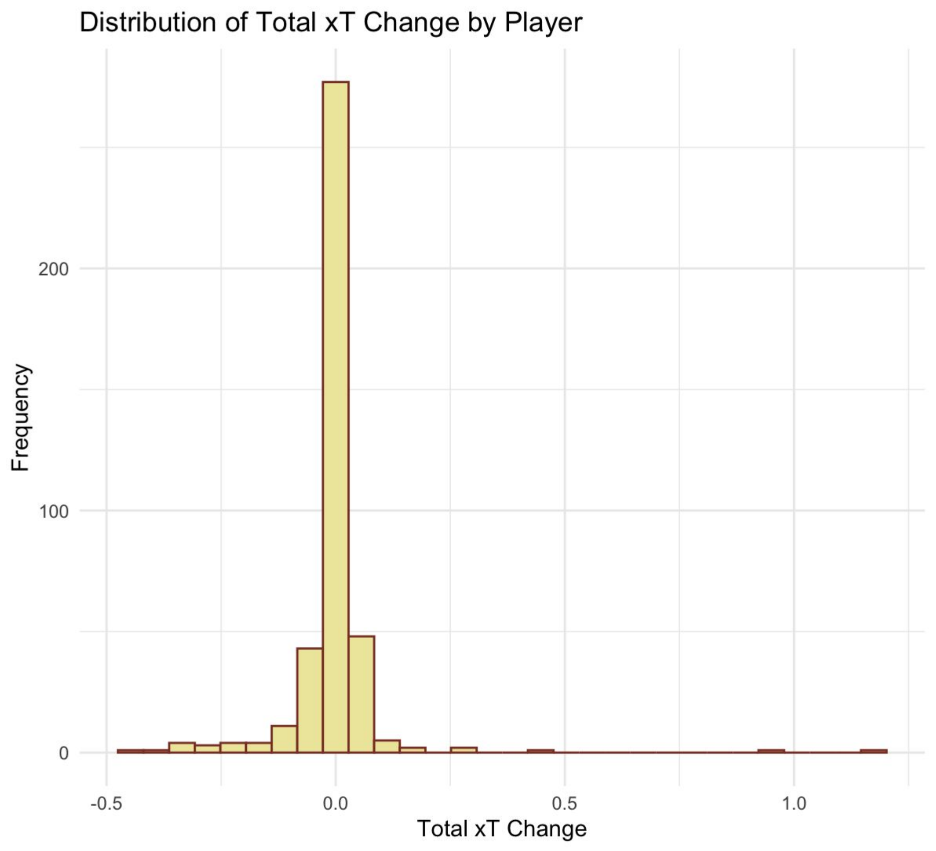


Number of Presses vs Average xT Change
Average xT Change
0.01
0.00
-0.01
25
50
75
100
Number of Presses



Number of Presses vs Total xT Change
Total xT Change
1.0
0.5
0.0
-0.5
25
50
75
100
Number of Presses



	pressurePlayer_nickname	total_xt	n_presses
1	Julian Alvarez	1.16889264781	67
2	Ollie Watkins	0.95161903596	77
3	Callum Wilson	0.46418764421	48
4	Mohamed Salah	0.28982481206	105
5	James Milner	0.27716533125	26
6	Leon Bailey	0.16912398822	64
7	Andreas Pereira	0.16254755388	76
8	Cristiano Ronaldo	0.10050019148	9
9	Liam Cooper	0.09471688590	13
10	Alexis Mac Allister	0.09434060530	61
11	Kenny Tete	0.09032806353	57
12	Harvey Elliott	0.08461184840	43
13	Thilo Kehrer	0.08379154679	44
14	Levi Colwill	0.08134769688	19
15	Fabinho	0.07180359149	53

Next Steps

- Finish KDE
- Bandwidth Estimation for KDE
- Evaluate metric against Wins, Goals Scored/Conceded

Thank you